

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

--

(Suggested maximum time: 10 minutes)

- (i) $781 + 27$

[illegible]

- [illegible]

- [illegible]

- | | | | |
|---------|-----------------|-------------------|----------------|
| $0.15,$ | $\frac{1}{20},$ | $\frac{17}{100},$ | $\frac{1}{10}$ |
|---------|-----------------|-------------------|----------------|

[illegible]

Answer = , , , .

- (i) What is the ratio of apples to pears to oranges?
Write your ratio in its simplest form.

[illegible]

Ratio: _____ : _____ : _____

- [illegible]

Question 2 (Suggested maximum time: 10 minutes)

Question 2 (Suggested maximum time: 10 minutes)

Kim lives in Belfast. She took part in a hockey tournament in France during September 2022. The tournament started on Wednesday, 14th September and finished on Saturday, 8th October.



- (a)** Work out how many days the tournament lasted.

Note: There are 30 days in September.

[illegible]

- (b)** Kim transferred £650 to her credit card to cover expenses in France.

The currency conversion rate at the time was €1 = £0.85.

Convert £650 to euro (€). Give your answer correct to the nearest cent.

[illegible]

- (c) The number of goals scored by Kim's team in each of the matches played are as follows:

4, 7, 2, 5, 6, 5, 2, 3, 2.

- (i) Is the data categorical or numerical?

Tick (✓) **one** box only. Give a reason for your answer.

Categorical

1

Numerical

1

Reason:

[illegible]

- (ii)** Work out the **range** of the goals scored by the team.

[illegible]

Question 3 (Suggested maximum time: 5 minutes)

A games console costs €305.

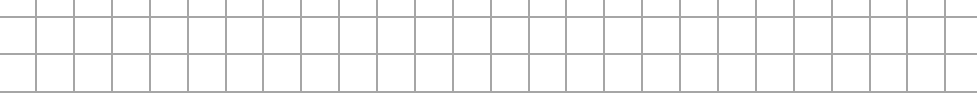
During a sale, the price of the games console is €244.



- (a) Work out the reduction in price of the games console during the sale.
- 

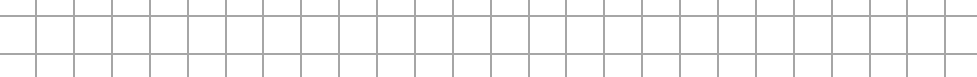
[illegible]

- (b)** Express this reduction as a percentage of the full price of the games console.



- (c) The shop also sells games for the games console.
Each game costs €45.
A bundle of three games is on sale, offering a saving of 25%.

Work out the cost of the three-game bundle.



Question 4 (Suggested maximum time: 5 minutes)

Question 4 (Suggested maximum time: 5 minutes)

The table below shows some values when 3 is raised to certain powers.

(a) Complete the table.

Power of 3	Expanded power of 3	Answer
3^1	3	3
3^2	3×3	
	$3 \times 3 \times 3$	27
3^4		81
3^5	$3 \times 3 \times 3 \times 3 \times 3$	
3^6		729

[illegible]

(b) Using the information in the table above, find the value of the following:

$$3^2 \times 3^4 =$$

--

(c) Without using a calculator and using the information in the table above, find the value of $\frac{729}{27}$. Show your workings out.

[illegible]

(Suggested maximum time: 10 minutes)

The graph shows a person's distance from the start over time. The y-axis is labeled 'Distance from start (km)' and ranges from 0 to 10. The x-axis is labeled 'Time (hours)' and ranges from 0 to 6. The graph consists of three segments: Segment A (0 to 2 hours, increasing from 0 to 6 km), Segment B (2 to 3 hours, constant at 6 km), and Segment C (3 to 4 hours, increasing from 6 to 9 km).

Time (hours)	Distance from start (km)
0	0
1	3
2	6
3	6
4	9

- Answer:

- [illegible]

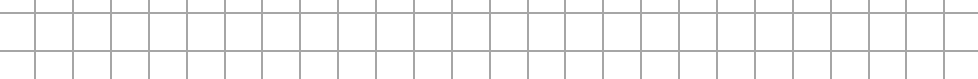
Answer:

[illegible]

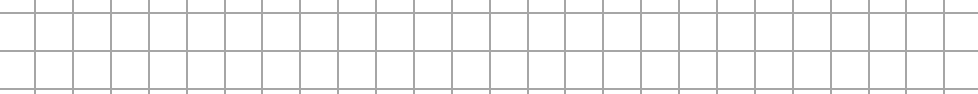
Question 6 (Suggested maximum time: 10 minutes)

Question 6 (Suggested maximum time: 10 minutes)

- (a)** Multiply out and simplify $(p - 2)(p + 5)$.



- (b)** Factorise fully $ab - 2b + 3a - 6$.



- (c)** Two expressions are shown in the boxes below:

$$6r - 5$$

$$4r + 7$$

- (i)** Find the value of each expression when $r = 5$.

$6r - 5 =$	$4r + 7 =$
------------	------------

- (ii) What value of r makes the two expressions equal?
Show your workings out.



Question 7 (Suggested maximum time: 5 minutes)

Question 7 (Suggested maximum time: 5 minutes)

Patrick receives a gift box containing 20 chocolates.

There are 10 milk chocolates and 7 dark chocolates in the box.

The remainder of the chocolates in the box are white.



- (a)** How many white chocolates are in the gift box?

[illegible]

- (b)** A chocolate is chosen at random from the gift box.
Write down the probability that the chocolate chosen is white.

Answer:

- (c)** The probability, that a particular type of chocolate is chosen at random from the box, is $\frac{1}{2}$.

What type of chocolate is chosen from the box?

Tick (✓) **one** box only.

milk chocolate

1

dark chocolate

11

white chocolate

1

[illegible]

Another box contains a selection of 32 dark and milk chocolates.

- (d)** Use the probability given in the table below to work out how many chocolates of each type are in this box. Fill in the missing values in the table.

Type	Probability	Number of chocolates
Milk	$\frac{5}{8}$	
Dark		

[illegible]

Question 8

(Suggested maximum time: 5 minutes)

Anna throws a six-sided die 50 times and records how many times it lands on each number, as shown in the table below.



- (a) Complete the table by calculating the number of rolls, probability or relative frequency for each side of Anna's die.

Side	Number of rolls	Probability	Relative Frequency
1	7	$\frac{7}{50}$	0.14
2	9	$\frac{9}{50}$	
3	10	$\frac{10}{50}$	
4		$\frac{6}{50}$	0.12
5			0.20
6			

[illegible]

- (b)** Anna says that she has used a fair die.

Do you agree with her?

Tick (✓) **one** box only. Give a reason for your answer.

Yes

11

No

7

Reason:

[illegible]

- (c)** What could Anna do to be more certain about whether or not her die is fair?

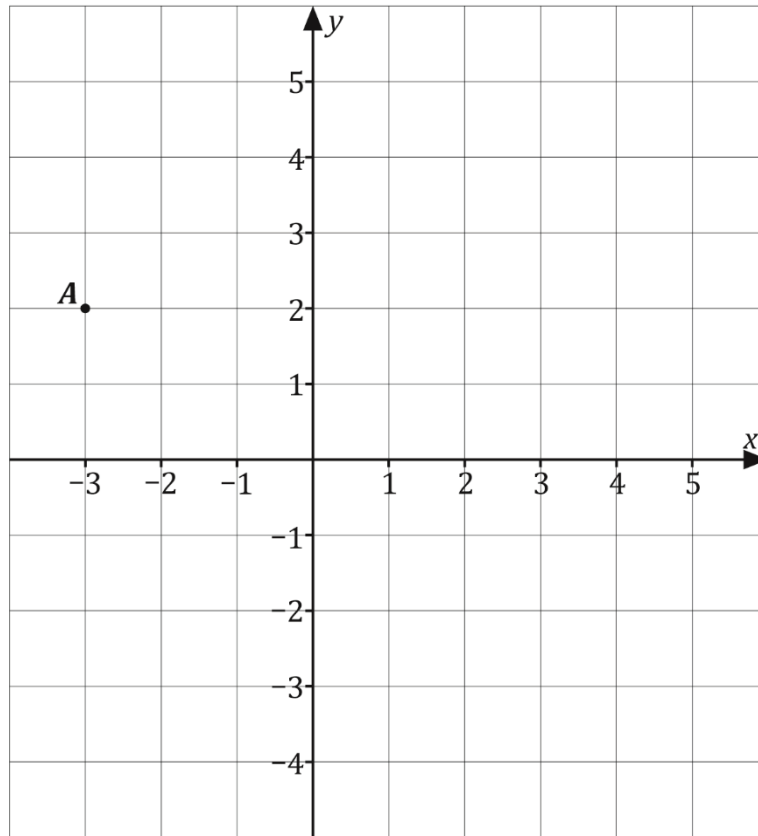
[illegible]

Question 9**(Suggested maximum time: 10 minutes)**

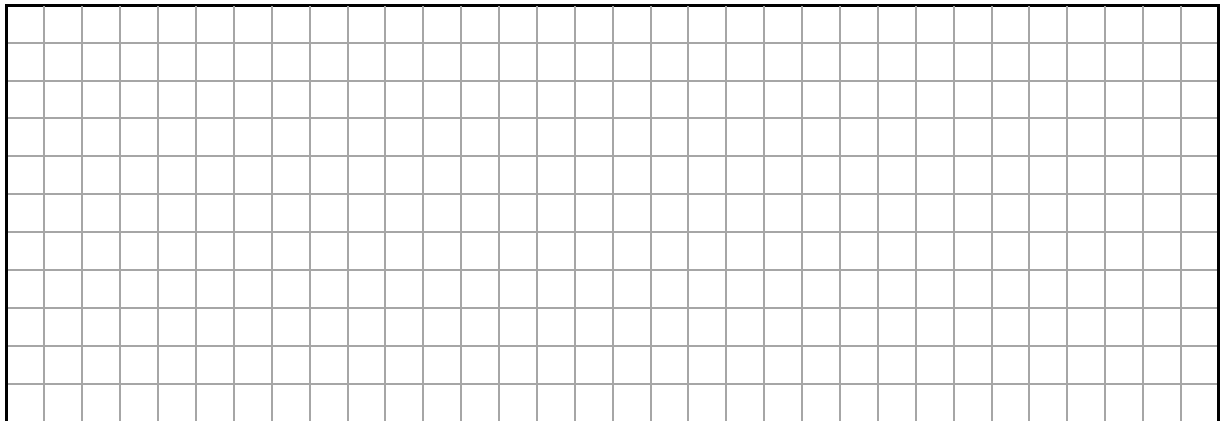
$A(-3, 2)$, $B(3, 4)$, $C(2, -3)$ and $D(1, 0)$ are four points.

(a) Plot the points above on the co-ordinate grid below.

Label the points clearly. The point A is plotted for you.



(b) Find the slope of the line AB .



- (c)** Using a straight edge (ruler), draw a line through the points A and B .
Draw another line through the points C and D until this line intersects the line AB .
- (d)** Write down the co-ordinates of the point where the line AB and the line CD intersect.
Label this point P on the diagram.

$$P = \begin{pmatrix} & \\ & \end{pmatrix}$$

- (e) Is $|AP| = |BP|$?
Tick (✓) **one** box only. Give a reason for your answer.

Yes

No

1

Reason: _____



Question 10

(Suggested maximum time: 10 minutes)

- (a) For each statement below, tick (✓) the correct box which makes the statement true for **right-angled triangles**. In each case, tick **one** box only.

- (i) The hypotenuse is the side opposite:

the acute angle

☐

the right angle

☐

any angle

☐

- (ii) The Theorem of Pythagoras can be used to find:

the area of the triangle

☐

any angle

☐

the length of any side

☐

- (iii) In trigonometry, the ratio that does **not** use the hypotenuse is:

cos

☐

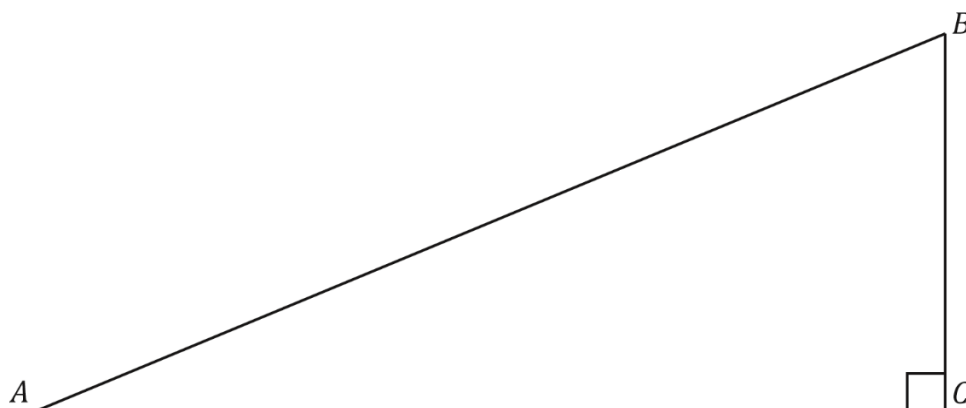
sin

☐

tan

☐

- (b) The diagram below shows a right-angled triangle ABC .



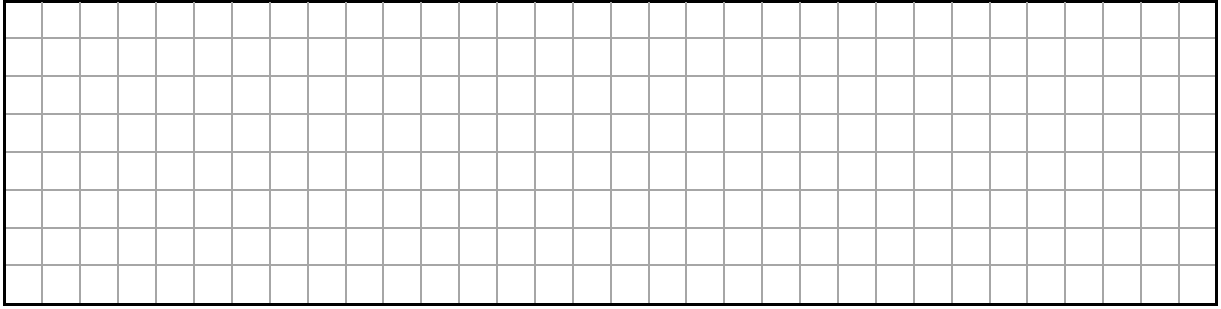
- (i) Use your ruler to measure, correct to the nearest cm, the lengths of the sides AB , BC and AC .

$|AB| =$ cm

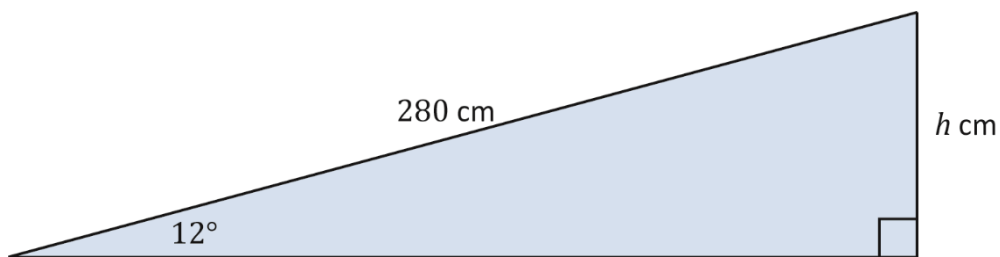
$|BC| =$ cm

$|AC| =$ cm

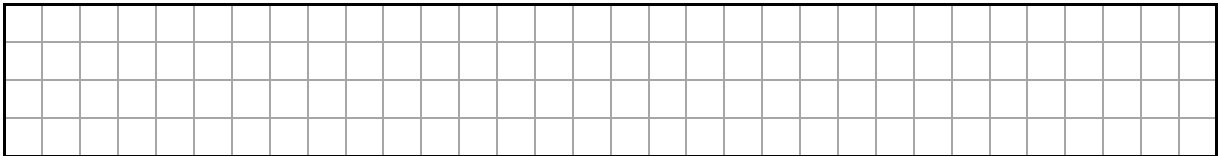
- (ii) Use the Theorem of Pythagoras to show that your answers in part (i) are correct.



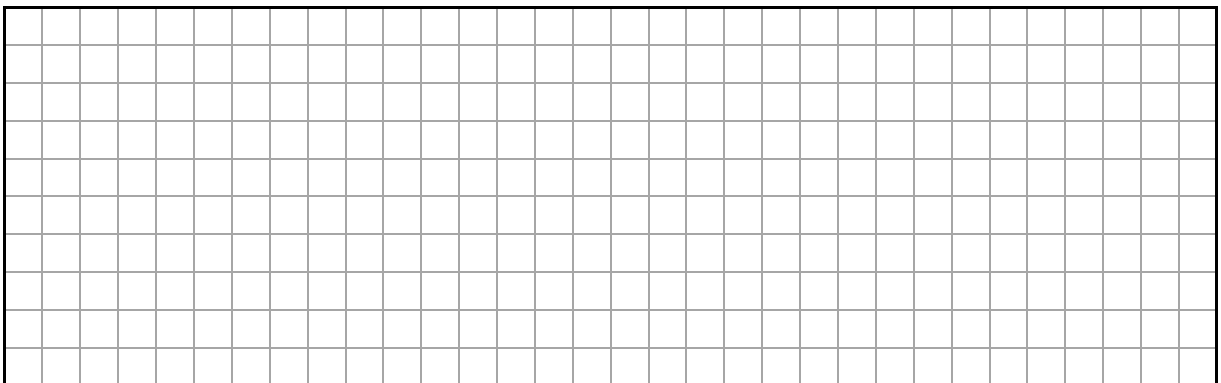
- (c) Tina designed a wheelchair access ramp for her local community centre. The ramp is 280 cm long and makes an angle of 12° with the horizontal, as shown in the diagram below (diagram not to scale).



- (i) Use a calculator to find the **value** of $\sin 12^\circ$, correct to one decimal place.



- (ii) Work out h , the height of the ramp, using a trigonometric ratio. Give your answer correct to the nearest cm.



Question 11**(Suggested maximum time: 10 minutes)**

(a) Complete the four statements below by using the following words in the boxes.

kilometres	grams	litres	centimetres
------------	-------	--------	-------------

(i) A jar of coffee can weigh 200

(ii) The length of a standard pencil is about 19

(iii) The distance from Dublin to Galway is about 207

(iv) A typical large bottle of sparkling water holds 2



-

- [illegible]

- [illegible]

- [illegible]

Question 12 (Suggested maximum time: 10 minutes)

Question 12 (Suggested maximum time: 10 minutes)

David recorded the county that each of the 45 students in his year-group had visited most often in the last year. He recorded his results and displayed them on a pie chart.

The table below shows his results.

County	Frequency	Angle of Sector
Dublin	10	80°
Galway	15	
Kerry	15	
Donegal		

- (a)** How many students have visited Donegal most often?

[illegible]

- (b)** How many degrees represents a full pie chart?

Tick (✓) **one** box only.

 180° 360°

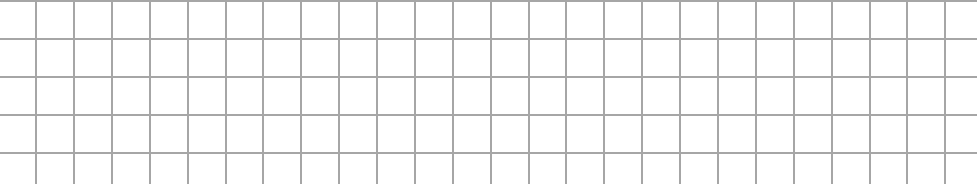
1

 90°

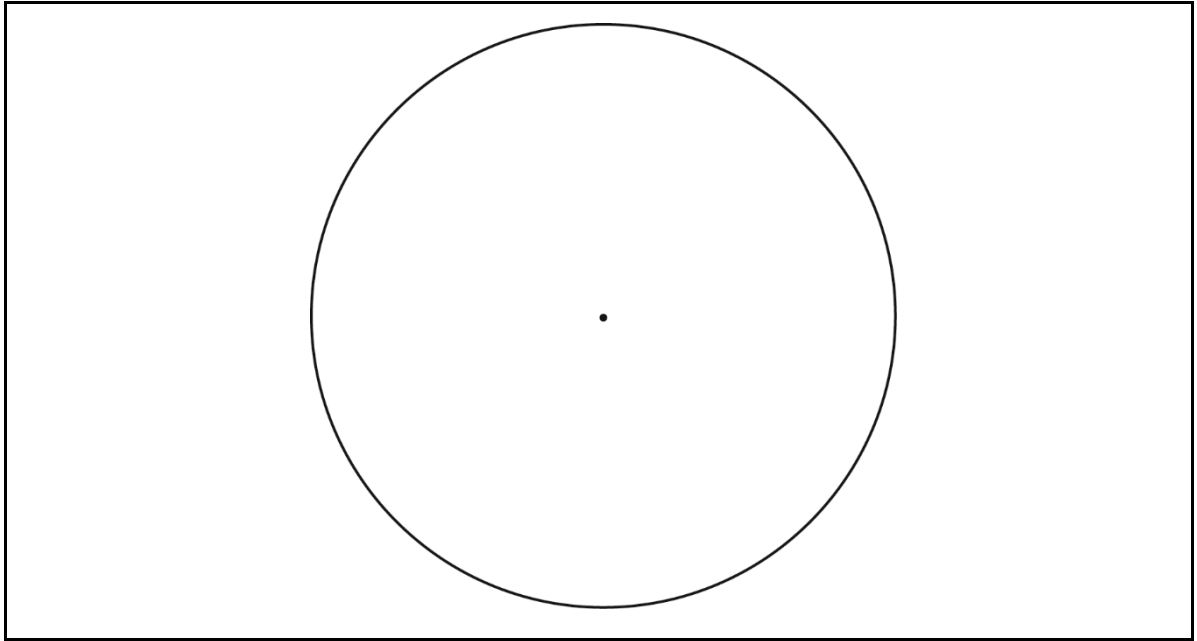
11

- (c)** Work out the sizes of the three missing angles for the pie chart.

Write each answer in the appropriate space in the table above.



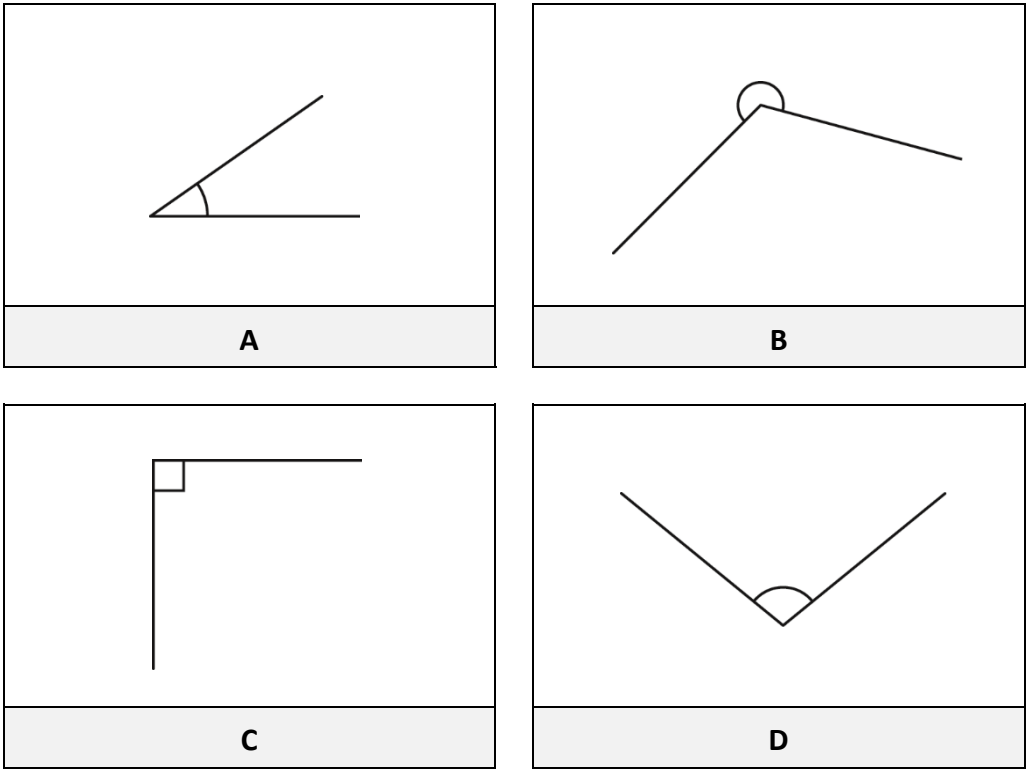
- (d) Complete the pie chart below to show the information in the table.
Clearly label each sector with the name of the county **and** the size of the angle.



Question 13

(Suggested maximum time: 5 minutes)

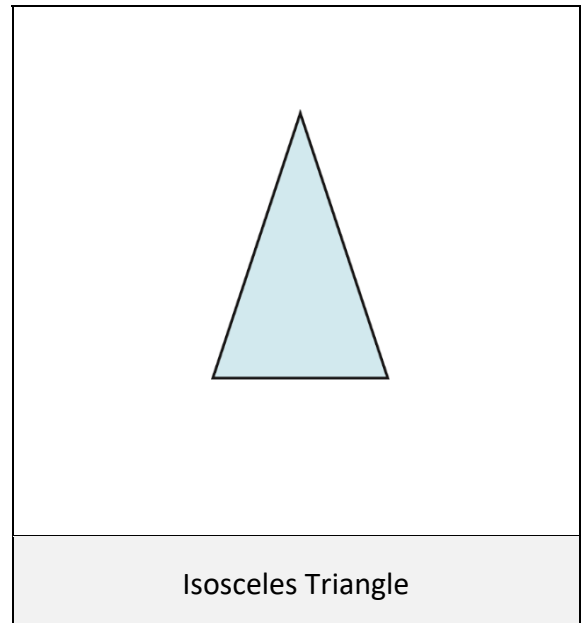
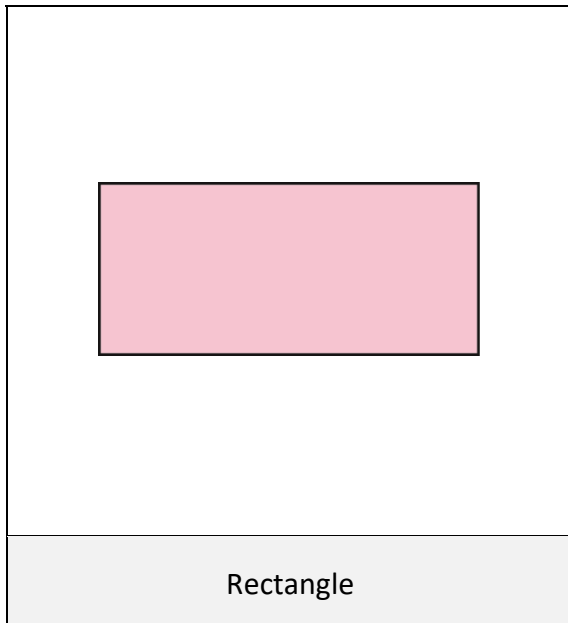
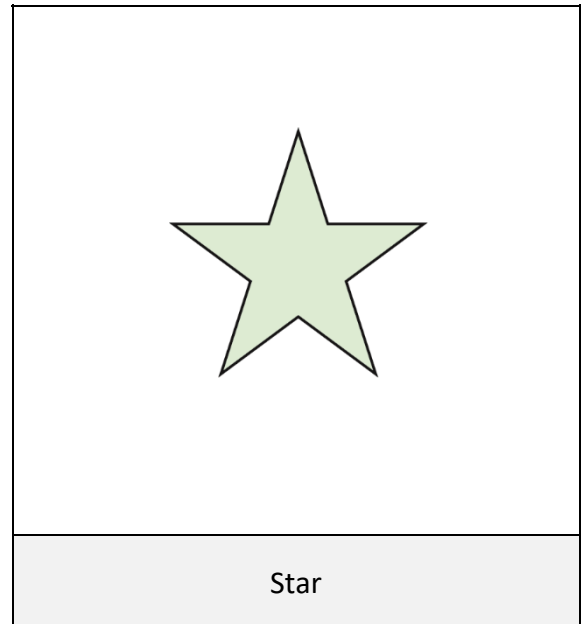
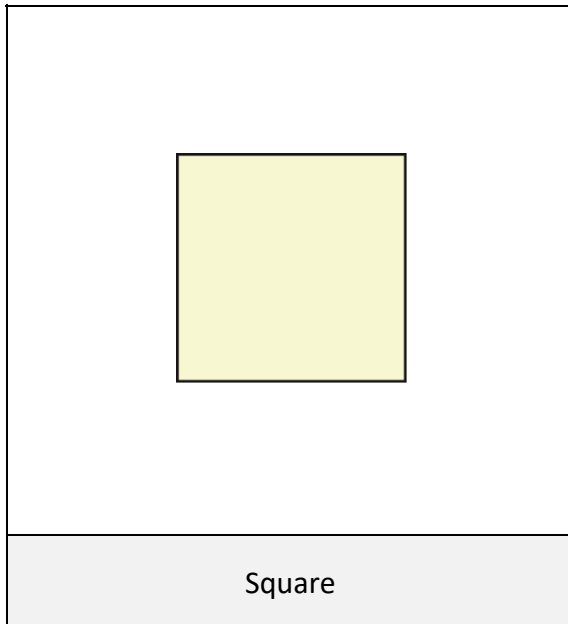
- (a) The diagrams below show different types of angles.
Each diagram is labelled **A**, **B**, **C** or **D**.



Insert **A**, **B**, **C** or **D** in table below to identify each type of angle shown in the diagrams above.

Type of angle	Letter
Acute	
Right-angle	
Reflex	
Obtuse	

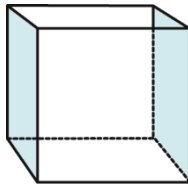
(b) Draw all of the axes of symmetry (if any) in each the following shapes.



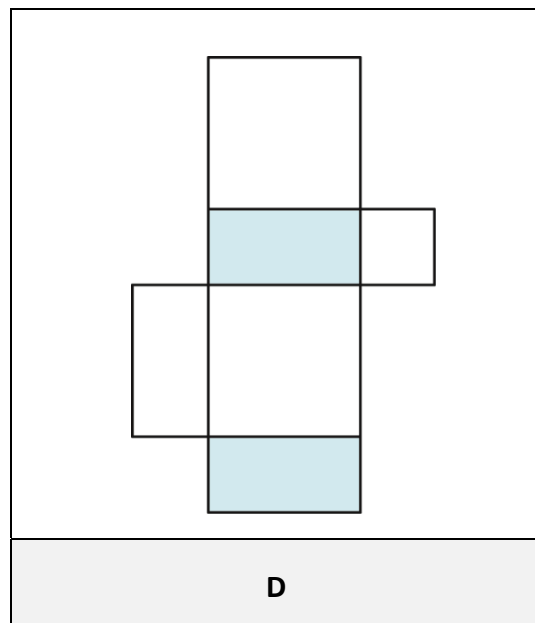
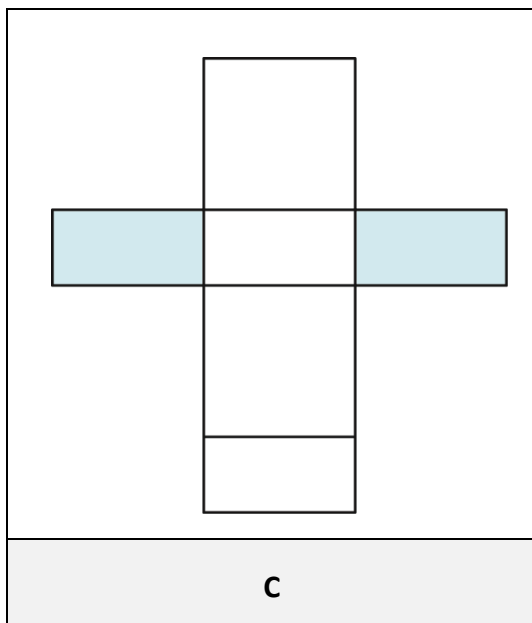
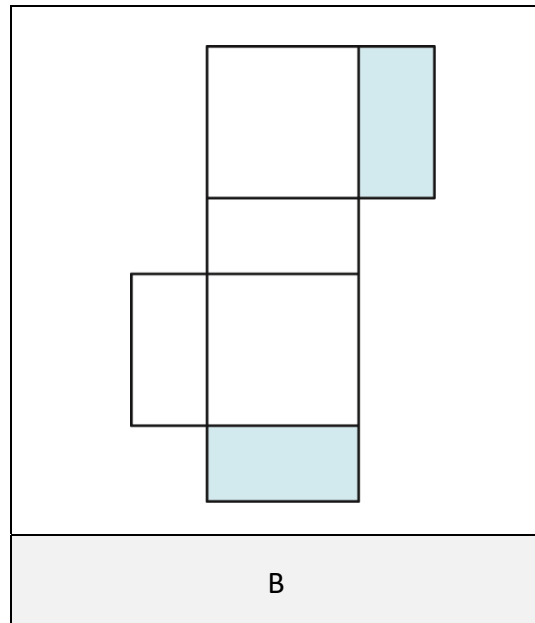
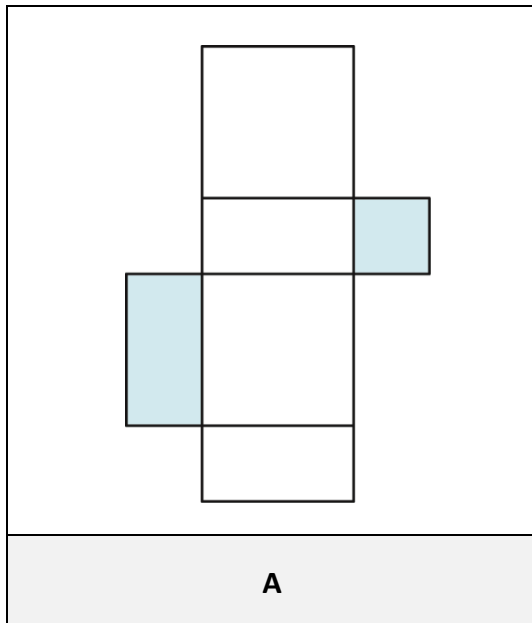
Question 14

(Suggested maximum time: 5 minutes)

- (a) The following diagram shows a rectangular prism.

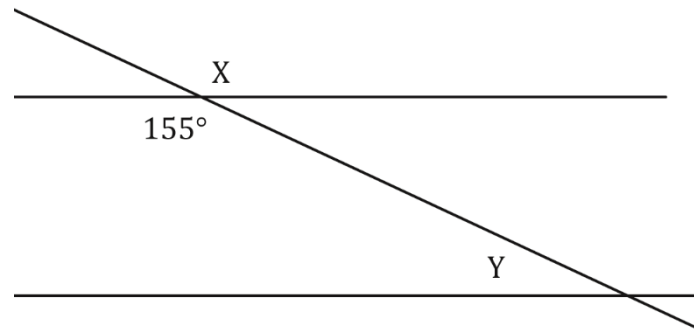


Which **one** of the following diagrams, **A**, **B**, **C** or **D** shows the correct net of the rectangular prism above.



Answer =

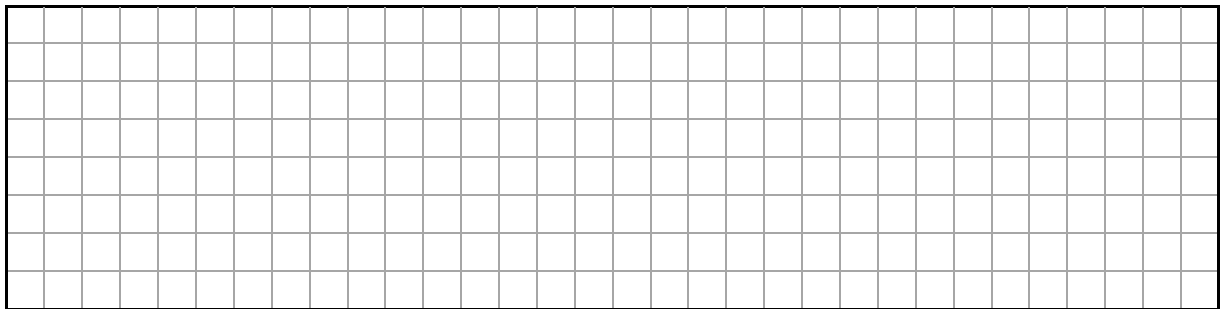
- (b) The diagram below shows two parallel lines and a transversal line.



- (i) Write down the value of the angle X .

$X =$ °

- (ii) Work out the value of the angle Y .



$Y =$ °

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Copyright notice

This examination paper may contain text or images for which DEB Exams is not the copyright owner, and which may have been adapted, for the purpose of assessment, without the authors' prior consent. This examination paper has been prepared in accordance with Section 53(5) of the *Copyright and Related Rights Act, 2000*. Any subsequent use for a purpose other than the intended purpose is not authorised. DEB Exams does not accept liability for any infringement of third-party rights arising from unauthorised distribution or use of this examination paper.

Pre-Junior Cycle Final Examination, 2023

Mathematics – Ordinary Level

Time: 2 hours

4e6b4069-6072-49fb-a557-f5acf7e96109



⑥ 2023-J019-1-EL-24/24